MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **WEEKLY TEST – 05** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. The value of $\dfrac{2\tan 30^\circ}{1-\tan^2 30^\circ}$ is equal to: [1]

(A) $\cos 60^\circ$ (B) $\sin 60^\circ$

(C) $\tan 60^\circ$ (D) $\sin 30^\circ$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $\tan 60^\circ$

Justification: $\dfrac{2(1/\sqrt{3})}{1-(1/\sqrt{3})^2} = \dfrac{2/\sqrt{3}}{2/3} = \dfrac{3}{\sqrt{3}} = \sqrt{3} = \tan 60^\circ.$

Q2. If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \dfrac{1}{\sqrt{3}}$, where $0^\circ < A+B \leq 90^\circ$ and $A > B$, then the value of A is: [1]

(A) 45° (B) 60° (C) 30° (D) 15°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 45°

Justification: $A+B = 60^\circ$ and $A-B = 30^\circ$. Adding gives $2A = 90^\circ \implies A = 45^\circ$.

Q3. In the previous question, the value of acute angle B is: [1]

(A) 15° (B) 30° (C) 45° (D) 60°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 15°

Justification: Substituting $A = 45^\circ$ into $A+B = 60^\circ$ gives $B = 60^\circ - 45^\circ = 15^\circ$.

Q4. The value of $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$ is: [1]

(A) 1 (B) 2 (C) 0 (D) $\sqrt{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 2

Justification: $2(1)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 = 2 + \frac{3}{4} - \frac{3}{4} = 2.$

Q5. **Assertion (A):** For any acute angle θ, $\sin^2\theta + \cos^2\theta = 1$.
Reason (R): $\sec^2\theta - \tan^2\theta = 1$ for all acute angles $0^\circ \le \theta < 90^\circ$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: Both (A) and (R) are true fundamental identities; however, (R) is an independent identity in $\sec\theta, \tan\theta$ (in fact derived by dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$) and does not explain (A).

SECTION B: Short Answer Type-I

MATHS MANTHAN | $(1 \times 2 = 2$ Marks)

Q6. If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = 1$, where $0^\circ < A+B \le 90^\circ$ and $A > B$, find the values of A and B. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Set Up Angle Equations [1 Mark]

$$\tan(A+B) = \sqrt{3} = \tan 60^\circ \implies A + B = 60^\circ \quad \text{— (i)}$$

$$\tan(A-B) = 1 = \tan 45^\circ \implies A - B = 45^\circ \quad \text{— (ii)}$$

Step 2: Solve for A and B [1 Mark]

Adding (i) and (ii): $2A = 105^\circ \implies A = 52.5^\circ$. Subtracting: $2B = 15^\circ \implies B = 7.5^\circ$.

$$\boxed{A = \mathbf{52.5^\circ}\ \left(\mathbf{52\tfrac{1}{2}^\circ}\right), \qquad B = \mathbf{7.5^\circ}\ \left(\mathbf{7\tfrac{1}{2}^\circ}\right)}$$

Q7. Prove that: $\dfrac{1+\cos\theta+\sin\theta}{1+\cos\theta-\sin\theta}=\dfrac{1+\sin\theta}{\cos\theta}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Multiply Numerator & Denominator by $(1+\cos\theta+\sin\theta)$ [1 Mark]

$$\text{LHS}=\frac{[(1+\cos\theta)+\sin\theta]^2}{(1+\cos\theta)^2-\sin^2\theta}$$

Step 2: Expand Both Using $\sin^2\theta+\cos^2\theta=1$ [1.5 Marks]

$$\text{Num}=(1+\cos\theta)^2+2\sin\theta(1+\cos\theta)+1-\cos^2\theta$$

$$=(1+\cos\theta)(1+\cos\theta+2\sin\theta+1-\cos\theta)=2(1+\cos\theta)(1+\sin\theta)$$

$$\text{Den}=1+2\cos\theta+\cos^2\theta-(1-\cos^2\theta)=2\cos\theta+2\cos^2\theta=2\cos\theta(1+\cos\theta)$$

Step 3: Cancel Common Factor $2(1+\cos\theta)$ [0.5 Mark]

$$\boxed{\text{LHS}=\frac{\mathbf{1}+\sin\theta}{\cos\theta}=\text{RHS}\quad[\text{Hence Proved}]}$$

Q8. Prove that: $\dfrac{\sin\theta}{1+\cos\theta}+\dfrac{1+\cos\theta}{\sin\theta}=2\csc\theta$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Combine Fractions over Common Denominator [1 Mark]

$$\text{LHS}=\frac{\sin^2\theta+(1+\cos\theta)^2}{(1+\cos\theta)\sin\theta}$$

Step 2: Expand & Simplify Using $\sin^2\theta+\cos^2\theta=1$ [1.5 Marks]

$$\text{LHS}=\frac{\sin^2\theta+1+2\cos\theta+\cos^2\theta}{(1+\cos\theta)\sin\theta}=\frac{2+2\cos\theta}{(1+\cos\theta)\sin\theta}=\frac{2(1+\cos\theta)}{(1+\cos\theta)\sin\theta}$$

Step 3: Result [0.5 Mark]

$$\boxed{\text{LHS}=\frac{\mathbf{2}}{\sin\theta}=\mathbf{2}\csc\theta=\text{RHS}\quad[\text{Hence Proved}]}$$

Q9. Prove that: $(\sec A - \tan A)^2 = \dfrac{1-\sin A}{1+\sin A}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Convert LHS into $\sin A$ and $\cos A$ [1 Mark]

$$\text{LHS} = \left(\frac{1}{\cos A} - \frac{\sin A}{\cos A}\right)^2 = \frac{(1-\sin A)^2}{\cos^2 A}$$

Step 2: Substitute $\cos^2 A = 1 - \sin^2 A$ & Cancel Factor [2 Marks]

$$\text{LHS} = \frac{(1-\sin A)^2}{(1-\sin A)(1+\sin A)} = \boxed{\frac{\mathbf{1}-\sin \mathbf{A}}{\mathbf{1}+\sin \mathbf{A}} = \text{RHS} \quad \text{[Hence Proved]}}$$

SECTION D: Case-Based / Applied Situation — MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. Satellite Earth Station Parabolic Antenna Alignment

A telecommunications ground station uses an adjustable parabolic dish antenna. A rigid calibration boom OP of length 4 m extends from the antenna vertex O to feed horn P. When tracking a geostationary satellite, the boom is inclined at an elevation angle of $\theta = 60°$ to the horizontal base line OQ. **Based on the above geometric model, answer the following questions:**

(i) Calculate the vertical height PQ of the feed horn above the horizontal base. **[1]**

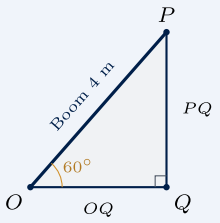
(ii) Calculate the horizontal distance OQ from the dish vertex to the base of the feed horn projection. **[1]**

(iii) Verify that $OP^2 = OQ^2 + PQ^2$ using the values obtained, and express $\tan 60°$ as the ratio $\dfrac{PQ}{OQ}$. **[2]**

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Sub-part (i): Vertical Height PQ [1 Mark]

$$PQ = OP\sin 60° = 4 \times \frac{\sqrt{3}}{2} = \boxed{\mathbf{2\sqrt{3}}\text{ m} \approx 3.46\text{ m}}$$



Sub-part (ii): Horizontal Distance OQ [1 Mark]

$$OQ = OP\cos 60° = 4 \times \frac{1}{2} \implies \boxed{OQ = \mathbf{2}\text{ m}}$$

Sub-part (iii): Pythagoras Verification & $\tan 60°$ [2 Marks]

$$OQ^2 + PQ^2 = 2^2 + (2\sqrt{3})^2 = 4 + 12 = 16 = 4^2 = OP^2 \quad \text{(Verified)}$$

$$\tan 60° = \frac{PQ}{OQ} = \frac{2\sqrt{3}}{2} \implies \boxed{\tan 60° = \sqrt{\mathbf{3}}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★